

| Input sequence: 3 − (−5) = |            |            |             | Predicted output: 8.02 |                |                 |                 |                 |                 |
|----------------------------|------------|------------|-------------|------------------------|----------------|-----------------|-----------------|-----------------|-----------------|
| Attention                  | 3<br>+0.10 | -<br>+0.06 | -5<br>+0.00 | 3 → -<br>+0.94         | - → 3<br>+0.90 | 3 → -5<br>+0.00 | -5 → 3<br>+0.00 | - → -5<br>+0.86 | -5 → -<br>+0.00 |
| Shapley values             | 3<br>+3.24 | -<br>+2.46 | -5<br>+2.34 |                        |                |                 |                 |                 |                 |
| Meta-Shapley values        | 3<br>+3.48 | -<br>-0.01 | -5<br>-0.32 | 3 → -<br>-0.22         | - → 3<br>-0.22 | 3 → -5<br>-0.02 | -5 → 3<br>-0.02 | - → -5<br>+2.69 | -5 → -<br>+2.69 |
| Shapley interactions       | 3<br>+3.48 | -<br>-0.01 | -5<br>-0.32 | 3 , -<br>-0.44         |                | 3 , -5<br>-0.04 |                 | - , -5<br>+5.37 |                 |
| Integrated gradients       | 3<br>+3.57 | -<br>+2.31 | -5<br>+2.16 |                        |                |                 |                 |                 |                 |
| Meta-Integrated gradients  | 3<br>+3.48 | -<br>-0.01 | -5<br>-0.32 | 3 → -<br>-0.54         | - → 3<br>+0.04 | 3 → -5<br>+0.02 | -5 → 3<br>+0.05 | - → -5<br>+2.47 | -5 → -<br>+2.86 |
| Integrated Hessians        | 3<br>+3.42 | -<br>+0.71 | -5<br>+0.55 | 3 , -<br>+0.06         | - , 3<br>+0.06 | 3 , -5<br>+0.09 | -5 , 3<br>+0.09 | - , -5<br>+1.52 | -5 , -<br>+1.52 |
| AttnLRP (≈ input×gradient) | 3<br>+0.45 | -<br>-1.61 | -5<br>+1.54 |                        |                |                 |                 |                 |                 |
| Meta-AttnLRP               | 3<br>+0.33 | -<br>-0.06 | -5<br>-0.59 | 3 → -<br>+0.08         | - → 3<br>+0.17 | 3 → -5<br>+0.07 | -5 → 3<br>-0.05 | - → -5<br>+2.06 | -5 → -<br>-1.62 |